Version 1.02.1

API Documentation

Setup

Ucurses framework setup information can be found in the Quick Start document file.

Namespaces

UCursesInclude
All UCurses classes are under this namespace.

Properties

UCurses Instance
Static property used to access the singleton object from any script in a scene.
Vector2Int GridSize
Property used to get the row and column size of the character grid.
Vector2Int DosScreenResolution
Property used to get the reference resolution the character grid uses.
float AspectRatio
Property used to get the aspect ratio of the displayed image used for the main camera.
Vector2Int CharacterSize
Property used to get the height and width of each character grid spot.
bool OffsetLine
Property used to get if the grid is using a horizontal offset line to increase the width of the character grid after each spot.
FilterMode ScreenFilterMode
Property used to get the texture filtering mode being used on the main camera image.
FilterMode CharacterFilterMode
Property used to get the texture filtering mode being used on each character space on the character grid.

Methods

void setDosScreenMode(DosScreenMode screenMode)		
Description	Sets the DOS console screen settings being emulated within Ucurses.	
Parameters	**DosScreenMode screenMode**	Takes a DosScreenMode container object with the required information needed to setup the character grid.

void setCharSprites(CharSprite[] charSprites)		
Description	Sets the array of ascii image sprites used for the character grid.	
Parameters	**CharSprite[] charSprites**	Takes a array of CharSprite container objects used for holding character sprites.

void printCharacter(Vector2Int position, char character, Color32 color)		
Description	Prints a character with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid where the character will be printed.
	char character	Character to be printed on the grid.
	Color32 color	The color of the printed character.

void printCharactersRectangle(Vector2Int position, char character, Color32 color, int width, int height)		
Description	Prints a rectangle of a character with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the top left most character of the rectangle being printed.
	char character	Character used in the rectangle to be printed on the grid.
	Color32 color	The color of the printed rectangle.
	int width	The width of the printed rectangle.
	int height	The height of the printed rectangle.

void printCharactersRectangle(Vector2Int position, char character, Color32 color, int width, int height, bool filled)		
Description	Prints a rectangle of a character with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the top left most character of the rectangle being printed.
	char character	Character used in the rectangle to be printed on the grid.
	Color32 color	The color of the printed rectangle.
	int width	The width of the printed rectangle.
	int height	The height of the printed rectangle.
	bool filled	If true the rectangle is filled, if false it is an outline.

void printString(Vector2Int position, string charString, Color32 color)		
Description	Prints a string of a characters with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the left most character of the string being printed.
	string charString	String to be printed on the grid.
	Color32 color	The color of the printed string.

void printNumberInt(Vector2Int position, int numericValue, Color32 color)		
Description	Prints a whole number with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the left most character of the number being printed.
	int numericValue	Int value to be printed to the grid.
	Color32 color	The color of the printed number.

void printNumberInt(Vector2Int position, int numericValue, Color32 color, uint paddingLength, bool addZeros)		
Description	Prints a whole number with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the top left most space of the rectangle being printed.
	int numericValue	Int value to be printed to the grid.
	Color32 color	The color of the printed number.
	uint paddingLength	How many padding spaces in front of the number are used.
	bool addZeros	If true adds zero characters in unused padding spaces.

void printNumberDouble(Vector2Int position, double numericValue, Color32 color)		
Description	Prints a decimal number with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the left most character of the decimal number being printed.
	double numericValue	Double value to be printed to the grid.
	Color32 color	The color of the printed decimal number.

void printNumberDouble(Vector2Int position, double numericValue, Color32 color, uint paddingLength, uint decimalLength, bool addZeros)		
Description	Prints a decimal number with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid of the left most character of the number being printed.
	double numericValue	Double value to be printed to the grid.
	Color32 color	The color of the printed number.
	uint paddingLength	How many padding spaces in front of the decimal place are used.
	uint decimalLength	How many padding spaces in back of the decimal place are used.
	bool addZeros	If true adds zero characters in unused padding spaces in front.

void printBackground(Vector2Int position, Color32 color)		
Description	Changes the background color of a character grid space.	
Parameters	Vector2Int position	Position on the grid where the background will be printed.
	Color32 color	The color of the printed background.

void printBackgroundRectangle(Vector2Int position, Color32 color, uint width, uint height)		
Description	Changes the background color of a rectangle of character grid spaces.	
Parameters	Vector2Int position	Position on the grid where the background will be printed.
	Color32 color	The color of the printed background.
	int width	The width of the rectangle being printed.
	int height	The height of the rectangle being printed.

void printBackgroundRectangle(Vector2Int position, Color32 color, int width, int height, bool filled)		
Description	Changes the background color of a rectangle of character grid spaces.	
Parameters	Vector2Int position	Position on the grid of the top left most space of the rectangle being printed.
	Color32 color	The color of the printed background.
	int width	The width of the rectangle being printed.
	int height	The height of the rectangle being printed.
	bool filled	If true the rectangle is filled, if false it is an outline.

Vector2Int mousePosition()	
Description	Retrieves the position of the mouse on the character grid.
Return	Returns what character grid position the mouse is currently over. If the mouse is outside the grid will return (-1, -1).

void characterBlink(Vector2Int position, bool enable)		
Description	Sets if character on the character grid is blinking.	
Parameters	Vector2Int position	Position on the grid where the character is/will be blinking. By default the blink is set to 0.5 seconds.
	bool enable	If true blinking is turned on, if false blinking is turned off.

void characterBlink(Vector2Int position, bool enable, float seconds)		
Description	Sets if character on the character grid is blinking.	
Parameters	Vector2Int position	Position on the grid where the character is/will be blinking.
	bool enable	If true blinking is turned on, if false blinking is turned off.
	float seconds	Sets the blink rate for the character in seconds.

void characterUnderlined(Vector2Int position, bool enable)		
Description	Sets if character on the character grid is underlined.	
Parameters	**Vector2Int position**	Position on the grid where the character is/will be underlined.
	bool enable	If true underline is turned on, if false underline is turned off.

void characterUnderlined(Vector2Int position, bool enable, uint width)		
Description	Sets if character on the character grid is blinking.	
Parameters	**Vector2Int position**	Position on the grid where the character is/will be underlined.
	bool enable	If true underline is turned on, if false underline is turned off.
	uint width	How grid spaces from the starting position the underline will go to the right.

string unixString(string stringValue)		
Description	Converts a string to use Unix style (LF) newline characters. All string based methods use this style.	
Parameters	**string stringValue**	The string to be converted.
Return	**string**	The converted string.

void printCharArray(Vector2Int position, char[,] charArray, Color32 color)		
Description	Prints a 2D array of characters with a given color on the character grid.	
Parameters	**Vector2Int position**	Position on the grid where the character is/will be underlined.
	char[,] charArray	The char array that you want to print to the string.
	Color32 color	The color of the printed char array.

void printCharArray(Vector2Int position, char[,] charArray, Color32[,] colorArray)		
Description	Prints a 2D array of characters with a corresponding 2D Color32 array on the character grid.	
Parameters	**Vector2Int position**	Position on the grid where the character is/will be underlined.
	char[,] charArray	The char array that you want to print to the string.
	Color32[,] colorArray	An array that holds the color for each char in the char array.

Change Log

1.0

Initial release.

1.01

Added multi-line support functionality for printString().

Added unixString() and printCharArray() methods.

1.02

Character sprites changed from individual files to a sprite map to improve performance.

1.02.1

Fixed bug with mousePosition() giving out of bounds coordinates if the mouse was on the edge of the screen.

Code Page 437 Character Set

Name	Char	ASCII
SOH (Start of Header)	☺	1
STX (Start of Text)	☻	2
ETX (End of Text)	♥	3
EOT (End of Transmission)	♦	4
ENQ (Enquiry)	♣	5
ACK (Acknowledge)	♠	6
BEL (Bell)	•	7
BS (BackSpace)	◘	8
HT (Horizontal Tabulation)	○	9
LF (Line Feed)	◙	10
VT (Vertical Tabulation)	♂	11
FF (Form Feed)	♀	12
CR (Carriage Return)	♪	13
SO (Shift Out)	♫	14
SI (Shift In)	☼	15
DLE (Data Link Escape)	►	16
DC1 (Device Control 1)	◄	17
DC2 (Device Control 2)	↕	18
DC3 (Device Control 3)	‼	19
DC4 (Device Control 4)	¶	20
NAK (Negative Acknowledge)	§	21
SYN (Synchronous Idle)	▬	22
ETB (End of Transmission Block)	↨	23
CAN (Cancel)	↑	24
EM (End of Medium)	↓	25
SUB (Substitute)	→	26
ESC (Escape)	←	27
FS (File Separator)	∟	28
GS (Group Separator)	↔	29
RS (Record Separator)	▲	30
US (Unit Separator)	▼	31
Space		32
Exclamation mark	!	33
Quotation Mark	"	34
Hash	#	35
Dollar	$	36
Percent	%	37
Ampersand	&	38
Apostrophe	'	39
Open bracket	(	40
Close bracket	)	41
Asterisk	*	42
Plus	+	43
Comma	,	44
Dash	-	45
Full stop	.	46

Name	Char	ASCII
Slash	/	47
Zero	0	48
One	1	49
Two	2	50
Three	3	51
Four	4	52
Five	5	53
Six	6	54
Seven	7	55
Eight	8	56
Nine	9	57
Colon	:	58
Semicolon	;	59
Less than	<	60
Equals sign	=	61
Greater than	>	62
Question mark	?	63
At	@	64
Upper case A	A	65
Upper case B	B	66
Upper case C	C	67
Upper case D	D	68
Upper case E	E	69
Upper case F	F	70
Upper case G	G	71
Upper case H	H	72
Upper case I	I	73
Upper case J	J	74
Upper case K	K	75
Upper case L	L	76
Upper case M	M	77
Upper case N	N	78
Upper case O	O	79
Upper case P	P	80
Upper case Q	Q	81
Upper case R	R	82
Upper case S	S	83
Upper case T	T	84
Upper case U	U	85
Upper case V	V	86
Upper case W	W	87
Upper case X	X	88
Upper case Y	Y	89
Upper case Z	Z	90
Open square bracket	[	91
Backslash	\	92

Name	Char	ASCII
Close square bracket	]	93
Caret	^	94
Underscore	_	95
Grave accent	`	96
Lower case a	a	97
Lower case b	b	98
Lower case c	c	99
Lower case d	d	100
Lower case e	e	101
Lower case f	f	102
Lower case g	g	103
Lower case h	h	104
Lower case i	i	105
Lower case j	j	106
Lower case k	k	107
Lower case l	l	108
Lower case m	m	109
Lower case n	n	110
Lower case o	o	111
Lower case p	p	112
Lower case q	q	113
Lower case r	r	114
Lower case s	s	115
Lower case t	t	116
Lower case u	u	117
Lower case v	v	118
Lower case w	w	119
Lower case x	x	120
Lower case y	y	121
Lower case z	z	122
Open brace	{	123
Pipe	|	124
Close brace	}	125
Tilde	~	126
Delete	⌂	127
Upper case C with cedilla	Ç	128
Lower case u with diaeresis	ü	129
Lower case e with acute	é	130
Lower case a with circumflex	â	131
Lower case a with diaeresis	ä	132
Lower case a with grave	à	133
Lower case a with ring above	å	134
Lower case c with cedilla	ç	135
Lower case e with circumflex	ê	136
Lower case e with diaeresis	ë	137
Lower case e with grave	è	138
Lower case i with diaeresis	ï	139
Lower case i with circumflex	î	140

Name	Char	ASCII
Lower case i with grave	ì	141
Upper case A with diaeresis	Ä	142
Upper case A with ring above	Å	143
Upper case E with acute	É	144
Lower case ae	æ	145
Upper case AE	Æ	146
Lower case o with circumflex	ô	147
Lower case o with diaeresis	ö	148
Lower case o with grave	ò	149
Lower case u with circumflex	û	150
Lower case u with grave	ù	151
Lower case y with diaeresis	ÿ	152
Upper case O with diaeresis	Ö	153
Upper case U with diaeresis	Ü	154
Cent sign	¢	155
Pound sign	£	156
Yen sign	¥	157
Peseta sign	₧	158
Lower case f with hook	ƒ	159
Lower case a with acute	á	160
Lower case i with acute	í	161
Lower case o with acute	ó	162
Lower case u with acute	ú	163
Lower case n with tilde	ñ	164
Upper case N with tilde	Ñ	165
Feminine ordinal indicator	ª	166
Masculine ordinal indicator	º	167
Inverted question mark	¿	168
Reversed not sign	⌐	169
Not sign	¬	170
Vulgar fraction one half	½	171
Vulgar fraction one quarter	¼	172
Inverted exclamation mark	¡	173
Left-pointing double angle quotation mark	«	174
Right-pointing double angle quotation mark	»	175
Light shade	░	176
Medium shade	▒	177
Dark shade	▓	178
Box drawings light vertical	│	179
Box drawings light vertical and left	┤	180
Box drawings vertical single and left double	╡	181
Box drawings vertical double and left single	╢	182
Box drawings down double and left single	╖	183
Box drawings down single and left double	╕	184
Box drawings double vertical and left	╣	185
Box drawings double vertical	║	186
Box drawings double down and left	╗	187
Box drawings double up and left	╝	188

Name	Char	ASCII
Box drawings up double and left single	╜	189
Box drawings up single and left double	╛	190
Box drawings light down and left	┐	191
Box drawings light up and right	└	192
Box drawings light up and horizontal	┴	193
Box drawings light down and horizontal	┬	194
Box drawings light vertical and right	├	195
Box drawings light horizontal	─	196
Box drawings light vertical and horizontal	┼	197
Box drawings vertical single and right double	╞	198
Box drawings vertical double and right single	╟	199
Box drawings double up and right	╚	200
Box drawings double down and right	╔	201
Box drawings double up and horizontal	╩	202
Box drawings double down and horizontal	╦	203
Box drawings double vertical and right	╠	204
Box drawings double horizontal	═	205
Box drawings double vertical and horizontal	╬	206
Box drawings up single and horizontal double	╧	207
Box drawings up double and horizontal single	╨	208
Box drawings down single and horizontal double	╤	209
Box drawings down double and horizontal single	╥	210
Box drawings up double and right single	╙	211
Box drawings up single and right double	╘	212
Box drawings down single and right double	╒	213
Box drawings down double and right single	╓	214
Box drawings vertical double and horizontal single	╫	215
Box drawings vertical single and horizontal double	╪	216
Box drawings light up and left	┘	217
Box drawings light down and right	┌	218
Full block	█	219
Lower half block	▄	220
Left half block	▌	221
Right half block	▐	222
Upper half block	▀	223
Greek lower case alpha	α	224
Lower case sharp s	ß	225
Greek upper case letter gamma	Γ	226
Greek lower case pi	π	227
Greek upper case letter sigma	Σ	228
Greek lower case sigma	σ	229
Micro sign	µ	230
Greek lower case tau	τ	231
Greek upper case letter phi	Φ	232
Greek upper case letter theta	Θ	233
Greek upper case letter omega	Ω	234
Greek lower case delta	δ	235
Infinity	∞	236
Greek lower case phi	φ	237
Greek lower case epsilon	ε	238
Intersection	∩	239
Identical to	≡	240
Plus-minus sign	±	241
Greater-than or equal to	≥	242
Less-than or equal to	≤	243
Top half integral	⌠	244
Bottom half integral	⌡	245
Division sign	÷	246
Almost equal to	≈	247
Degree sign	°	248
Bullet operator	∙	249
Middle dot	·	250
Square root	√	251
Superscript lower case n	^{n}	252
Superscript two	^{2}	253
Black square	■	254